Arrival · Supported

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Fraction

What percentage is 1/4?

Help: Read or point first. Multiply by 100.

2. Genotypes

What is the genotype ratio for Aa × Aa?

Help: Read or point first. Count AA, Aa and aa.

3. Phenotypes

With complete dominance, what is the phenotype ratio?

Help: Read or point first. AA and Aa show the dominant phenotype.

4. Independence

Does an aa offspring force the next offspring to be AA?

Help: Read or point first. Each fertilisation is a new event.

My response: _____

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Independent · Supported

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Calculate

Complete the expected aa counts for A, B and C.

Help: For each sample, write total ___. Find one quarter of that total for the expected aa count.

2. Convert

Convert observed 7/20 and 18/80 to percentages.

Help: Write two calculations: 7 ÷ 20 × 100 and 18 ÷ 80 × 100. Keep the correct total each time.

3. Explain

Explain why 7/20 does not automatically disprove the one-in-four model.

Help: Compare predicted probability with this small sample. Use: expected ___, observed ___; chance can produce ___.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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Exit ticket · Supported

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

1. Calculate

Cross Aa × Aa. Find the expected aa count in 80 offspring.

Help: Find one quarter of 80. Write: working ___; answer ___; unit or percent ___.

2. Interpret

Why might a sample of 20 contain 7 aa rather than exactly 5?

Help: Use: my answer is ___ because ___. Point to a model or card first. Small samples vary by chance.

My response: _____

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Arrival · Independent

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Fraction

What percentage is 1/4?

Help: Multiply by 100.

2. Genotypes

What is the genotype ratio for Aa × Aa?

Help: Count AA, Aa and aa.

3. Phenotypes

With complete dominance, what is the phenotype ratio?

Help: AA and Aa show the dominant phenotype.

4. Independence

Does an aa offspring force the next offspring to be AA?

Help: Each fertilisation is a new event.

My response: _____

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Independent · Independent

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Calculate

Complete the expected aa counts for A, B and C.

Help: Total × 0.25.

2. Convert

Convert observed 7/20 and 18/80 to percentages.

Help: Count ÷ total × 100.

3. Explain

Explain why 7/20 does not automatically disprove the one-in-four model.

Help: Expected proportions need not match small samples exactly.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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Exit ticket · Independent

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

1. Calculate

Cross Aa × Aa. Find the expected aa count in 80 offspring.

Help: Find one quarter of 80.

2. Interpret

Why might a sample of 20 contain 7 aa rather than exactly 5?

Help: Small samples vary by chance.

My response: _____

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Arrival · Aspire

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Fraction

What percentage is 1/4?

Help: Multiply by 100.

2. Genotypes

What is the genotype ratio for Aa × Aa?

Help: Count AA, Aa and aa.

3. Phenotypes

With complete dominance, what is the phenotype ratio?

Help: AA and Aa show the dominant phenotype.

4. Independence

Does an aa offspring force the next offspring to be AA? Give a reason.

Help: Each fertilisation is a new event.

My response: _____

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Independent · Aspire

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Calculate

Complete the expected aa counts for A, B and C.

Help: Total × 0.25.

2. Convert

Convert observed 7/20 and 18/80 to percentages.

Help: Count ÷ total × 100.

3. Keep events independent

Three offspring are not aa. What is the predicted chance that the next is aa in this model? Explain.

Help: Do not subtract previous outcomes from the four-box diagram.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

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Exit ticket · Aspire

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

1. Calculate

Cross Aa × Aa. Find the expected aa count in 80 offspring.

Help: Find one quarter of 80.

2. Keep events independent

Three offspring are not aa. What is the predicted chance that the next is aa in this model? Explain.

Help: Do not subtract previous outcomes from the four-box diagram.

My response: _____

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Word page

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Word	Meaning
Allele	A version of a gene.
Genotype	The allele pair for a gene.
Phenotype	An observable trait.
Dominant	Shown with one copy in this simple model.
Recessive	Shown with two recessive copies in this model.
Gamete	A sex cell with one allele for each gene in this model.
Probability	The chance of an outcome.
Expected	What the model predicts on average.
Observed	What was counted in a sample.

My language: _____

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Knowledge organiser

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Convert genetic probabilities between fractions, ratios and percentages, and explain why observations vary.

Aa × Aa · side / top	**A**	**a**
A	AA	Aa
a	Aa	aa

AA : Aa : aa is 1 : 2 : 1. With complete dominance, the phenotype ratio is 3 : 1. The aa chance is 1/4 = 25%.

Chance × total gives an expected count. Count ÷ total × 100 gives an observed percentage.

This is a simple model for one gene. A dominant allele is not stronger, better or more common. Each fertilisation is a new chance. Do not use real family traits or ask for personal or medical details.

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Starter

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Aa × Aa predicts a 25% chance of aa. Which statement follows?

A. Exactly one in every family of four must be aa

B. Each fertilisation has a one-in-four chance under this model

C. After three non-aa offspring, aa is certain

Think first. Choose and give one science reason.

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Describe the parents

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Parent 1	Parent 2
Aa	Aa

Both parents are Aa. Each has one A and one a allele. This one-gene model uses complete dominance.

This is a simple model for one gene. Choose when to move to the next step.

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Separate into gametes

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Aa × Aa · side / top	**A**	**a**
A	_____	_____
a	_____	_____

Each parent can produce gametes carrying A or a. Put one allele in each margin box.

This is a simple model for one gene. Choose when to move to the next step.

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Combine one from each

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Aa × Aa · side / top	**A**	**a**
A	AA	Aa
a	Aa	aa

Each offspring box combines one allele from the top with one from the side: AA, Aa, Aa and aa.

This is a simple model for one gene. Choose when to move to the next step.

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Read the probability

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Aa × Aa · side / top	**A**	**a**
A	AA	Aa
a	Aa	aa

The aa chance is one in four (25%) each time. Four offspring do not have to match this pattern.

This is a simple model for one gene. Choose when to move to the next step.

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We do · sort using the evidence

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Choose a group for each card. Give the clue for one choice. You can point or direct an adult.

Card	Your group
Model: 5 aa in 20	_____
Model: dominant trait in 75%	_____
Sample: 7 aa counted	_____
Next sample: 3 aa counted	_____
Four offspring must give one aa	_____
The next one must make the counts fit	_____

Groups: Model prediction / Sample count / False claim.

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

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I do 2 · worked example

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

A model gives a 50% chance of an outcome. Find the expected count in 30 trials.

Write 50% as 0.5.

Expected count = 0.5 × 30 = 15.

The model predicts 15. A real count can be more or less than 15.

Each trial has its own chance. Its result does not fix the next one.

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Expected versus observed

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

All samples model Aa × Aa. Calculate the expected aa count, then compare it with the simulated result. Do not change the theoretical probability because of one sample.

Sample	Total	Expected aa	Observed aa
A	20	___	7
B	20	___	3
C	80	___	18

My evidence and reasoning

One limit or useful check

My response: _____

Keep your first answer. Hear the feedback. Choose one real change, or explain why to keep it. Ask for a read-back if it helps.

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Punnett square · working sheet

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Record the parent alleles. Fill all four boxes. Then find the chance of aa.

Aa × Aa · side / top	**A**	**a**
A	_____	_____
a	_____	_____

Compare with Aa × aa.

Aa × aa · side / top	**A**	**a**
a	_____	_____
a	_____	_____

Aa × Aa chance of aa: _____ . Aa × aa chance of aa: _____ .

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Evidence cards

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Cut apart, point, or direct an adult. Match a card and its clue. Say why.

Card	Clue
Model: 5 aa in 20	20 × 1/4 = 5.
Model: dominant trait in 75%	Three of the four boxes show this trait.
Sample: 7 aa counted	This count comes from one trial of the model.
Next sample: 3 aa counted	Samples can differ by chance.
Four offspring must give one aa	The model gives a chance. It does not force a set order.
The next one must make the counts fit	Each event has its own chance. Earlier results do not force the next one.

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